Let R, S be rings.

$$\mathrm{Hom}(R,S) = \{\varphi : R \to S \mid \varphi \text{ is ring-Homomorphism}\}.$$

And, $\mathrm{Emb}(R,S) = \{\varphi \in \mathrm{Hom}(R,S) \mid \varphi \text{ is } 1\text{-}1\}$.

Lemma. Let E/F be an algebraic extension, and L be a field. Let $\varphi: F \to L$ be a field embedding. Given $\alpha \in E \setminus F$, a map

$$\varphi': F(\alpha) \to L(\beta): \sum a_i \alpha^i \mapsto \sum \varphi(a_i) \beta^i$$

is an extension embedding of φ, where β is a root of m_α^φ.

Proof.

Thm. Let F be a field, and E be an algebraic extension of F. Let $\varphi: F \to L$ be an embedding of F into an algebraically closed field L. Then, there exists an extension of φ to an embedding of E in L.

Moreover, if E is algebraically closed and L is algebraic over $\varphi[E]$, then any extension of φ to E is an isomorphism of E into L.

Proof. Define $S := \{(K, \tau) \mid F \le K \le E,\ \tau: K \hookrightarrow L \text{ such that } \tau|_F = \varphi\}$

Since $(F, \varphi) \in S$, $S \neq \emptyset$. And, define an order in S as:

$$(K_1, \tau_1) \le (K_2, \tau_2) \iff K_1 \subseteq K_2 \text{ and } \tau_2|_{K_1} = \tau_1.$$

Then, $(S, \le)$ is a partially-ordered set, because: reflexivity and anti-symmetry are clear, and transivity is given by: if $(K_1, \tau_1) \le (K_2, \tau_2) \le (K_3, \tau_3)$, then, $K_1 \subseteq K_3$ is clear and since $\tau_3|_{K_2} = \tau_2$ and $\tau_2|_{K_1} = \tau_1$, so

for all $a \in K_1$, $\tau_3(a) = \tau_2(a) = \tau_1(a)$, so $\tau_3|_{K_1} = \tau_1$.

Let $\{(K_i, \tau_i) \mid i \in I\} \subseteq S$ be a totally-ordered subset. Define

$$K := \bigcup_{i \in I} K_i \quad \text{and} \quad \tau: K \to L;\ a \mapsto \tau_j(a) \text{ where } j \in I \text{ such that } a \in K_j.$$

Such τ is well-defined, because: given $a \in K$, let $j_1, j_2 \in \{j \in I \mid a \in K_j\}$. Since $\{(K_i, \tau_i) \mid i \in I\}$ is totally ordered, so $K_{j_1} \subseteq K_{j_2}$ or $K_{j_2} \subseteq K_{j_1}$. If $K_{j_1} \subseteq K_{j_2}$, then $\tau_{j_2}(a) = \tau_{j_2}|_{K_{j_1}}(a) = \tau_{j_1}(a)$. Similarly, the case of $K_{j_2} \subseteq K_{j_1}$ is same. Now, (K, τ) is an upper bound for the totally ordered subset, and $(K, \tau) \in S$. By the Zorn's lemma, there exists a maximal element $(M, \Phi) \in S$. To show $M = E$, assume $\alpha \in E$ but $\alpha \notin M$. Since E is algebraic over F, so is over M and therefore $M(\alpha)$ is a finite extension of M, therefore there is an extension of Φ as

$$\Phi: M(\alpha) \to L;\ \sum a_i \alpha^i \mapsto \sum \Phi(a_i) \cdot \beta^i$$

where $\beta \in L$ is a root of evaluate of a minimal polynomial of α. But, it contradicts to the maximality.

Thm. Let K/F be a splitting field of a polynomial $f(x) \in F[x]$
If E is another splitting field of f, then there exists an isomorphism

$$\sigma : E \to K \text{ fixing } F$$

And, if $F \leq K \leq \overline{K}$, then every embedding $\varphi : E \to \overline{K}$ fixing F must be an isomorphism of E onto K.

Proof. By the extension theorem, the identity map on F

$$id : F \to \overline{K} ; a \mapsto a$$

can be extended as $\sigma : E \to \overline{K}$. Over E, we have a factorization

$$f(x) = c \cdot (x - \beta_1) \cdots (x - \beta_n) \text{ where } \beta_1, \cdots, \beta_n \in E, \ c \in F.$$

Since σ fixes F, so the induced map $f \mapsto f^\sigma$ fixes f, so that

$$f(x) = f^\sigma(x) = c \cdot (x - \sigma(\beta_1)) \cdots (x - \sigma(\beta_n))$$

and, $f(x)$ can be factorized over K as

$$f(x) = c \cdot (x - \alpha_1) \cdots (x - \alpha_n) \text{ where } \alpha_1, \cdots, \alpha_n \in E$$

and since $\overline{K}[x]$ is a U.F.D., it follows that the tuple $(\sigma(\beta_1), \cdots, \sigma(\beta_n))$ differs from $(\alpha_1, \cdots, \alpha_n)$ by a permutation.
Now, we conclude

$$K = F(\alpha_1, \cdots, \alpha_n) = F(\sigma(\beta_1), \cdots, \sigma(\beta_n))$$

and so that $\sigma[E] = K$.